VANISHING OF THE NORMALIZATION OF I_{M!}U ABOVE DEGREE M

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: rides on registered Lemma lemma-simplicial-abelian-n-skel-sub; the normalization N(V') is trivial above degree m by that lemma, so V = V' gives N(V)_n = 0 for n > m.

Editorial note. The label lemma-imshriek-abelian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Normalized subobject. Same unit, Lemma lemma-simplicial-abelian-n-skel-sub. The subobject V' with trivial normalization in high degrees is the one of Lemma lemma-simplicial-abelian-n-skel-sub.

Source reference. Original reference 017Y: sources/stacks-simplicial/pinned-original-source.tex, source lines 2310–2310. Prerequisite pointer invoked at proof line(s) 3385; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category Let U be an m-truncated simplicial object of $\mathcal{A}$. For $n > m$ we have $N(i_{m!}U)_n = 0$.*

Proof. Write $V = i_{m!}U$. Let $V' \subset V$ be the simplicial subobject of V with $V'_i = V_i$ for $i \leq m$ and $N(V'_i) = 0$ for $i > m$, see Lemma 017Y. By the adjunction formula, since $\mathrm{sk}_m V' = U$, there is an inverse to the injection $V' \to V$. Hence $V' = V$. □

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.